

Properties of Bulk matter.

Elasticity.

Deforming force: when some force is applied on a body and due to it, either the size or the shape of the body changes, the force is said to be a deforming force.

Elastic body: a body is said to be an elastic body if it regains its original size or shape after the removal of deforming force.

e.g; trampoline, spring.

Plastic body: a body is said to be a plastic body if it doesn't regain its original configuration (size or shape) after the removal of deforming force.

e.g; wax, mud.

Restoring force: it is the internal force developed in a body due to which the body regains its original configuration. In an elastic body,

$$\text{restoring force} = \text{deforming force}.$$

Elastic after effect: when deforming force is removed from an elastic body, sometimes the elastic body takes time to regain its original configuration. This delay in regaining the original configuration after the removal of deforming force is known as elastic after effect.

Elasticity is defined as the property by the virtue of which an elastic body regains its original configuration after the removal of deforming force.

Stress: developed in a body is defined as deforming force per unit area.

$$\text{stress} = \frac{\text{deforming force}}{\text{area}}$$

In case of elastic body.

Deforming force = Restoring force.

$$\text{stress} = \frac{\text{restoring force}}{\text{Area}}$$

SI unit of stress = N/m^2 .

$$\text{DF} = \frac{\text{M}^1 \text{L}^1 \text{T}^{-2}}{\text{L}^2} = \text{M}^1 \text{L}^{-1} \text{T}^{-2}$$

Types of stress

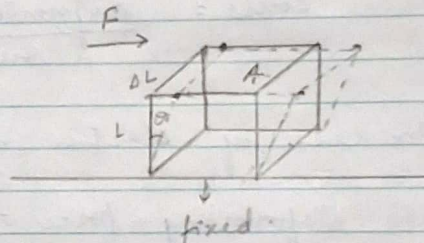
1 Longitudinal stress / normal stress: It is the stress (restoring force per unit area) developed in a body when the length of the body changes. It is of two types:

- a) Tensile stress: It is the stress developed in the body when its length increases
- b) compressive stress: It is the stress developed in a body when its length decreases.

2 volumetric stress: It is the stress developed

in a body when the volume of the body changes.

- 3 Tangential stress / shearing stress: It is the stress developed in a body when some deforming force is applied tangentially to it due to which its shape changes.



STRAIN.

$$= \frac{\text{change in configuration}}{\text{original configuration}}$$

strain is unitless as well as dimensionless.

Types of strain:

- 1 Longitudinal strain: $\frac{\text{change in length}}{\text{original length}}$

$$L.S = \frac{\pm \Delta L}{L}$$

- 2 Hydraulic strain / volumetric strain:

$$\frac{\text{change in volume}}{\text{original volume}}$$

$$= \frac{\pm \Delta V}{V}$$

3

Tangential strain / shear strain:

$$\theta = \frac{\text{change in shape}}{\text{original shape.}}$$

$$\theta = \frac{\Delta L}{L}$$

$$\theta = \frac{\Delta r}{R}$$

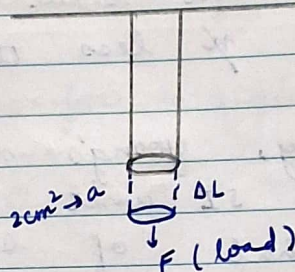
Hook's Law

It states that the extension produced in a body is directly proportional to the load applied to it.

$$F \propto \Delta L$$

$$\frac{F}{a} \propto \frac{\Delta L}{L}$$

Stress $\propto$ strain



Another statement.

It states that within elastic limit, the stress produced in a body is directly proportional to the strain produced in it.

stress $\propto$ strain.

$$\boxed{\text{Stress} = E \times \text{strain}} \quad \text{--- Hooke's law.}$$

where E is a constant and is known as coefficient of elasticity / modulus of elasticity.

$$E = \frac{\text{Stress}}{\text{Strain}}$$

SI unit and dimensional formula of coefficient of elasticity are same as that of stress.
 $(\text{N/m}^2, \text{M}^1 \text{L}^{-1} \text{T}^{-2})$.

- Types of elasticity.
- 1) Young's modulus of elasticity (γ)
- $$\gamma = \frac{\text{longitudinal stress}}{\text{longitudinal strain}}$$

$$\gamma = \frac{F/a}{\Delta L/L} = \frac{FL}{a\Delta L}$$

Young's modulus of elasticity is involved in solids only.

more the value of γ , more elastic is the body & less the γ , less elastic is the body.

clearly, young's modulus of elasticity has same SI unit and dimensional formula as that of stress (N/m^2 , $\text{ML}^{-1}\text{T}^{-2}$)

- 2) Bulk modulus of elasticity (B):

$$B = \frac{\text{volumetric stress}}{\text{volumetric strain}}$$

$$B = \frac{F/a}{\Delta V/V} = \frac{FV}{a\Delta V}$$

Bulk modulus of elasticity is involved in solids, liquids as well as gases

clearly, it is ^{has} same SI unit and DF as that of strain.

∴ Imp: compressibility (C)

$$= \frac{1}{\text{Bulk modulus}}$$

$$B = \frac{1}{C}$$

As $C_g > C_l > C_s$

$$\therefore B_g < B_l < B_s$$

3) Modulus of rigidity / shear modulus of elasticity (G)

$$G = \frac{\text{shearing stress}}{\text{shearing strain}}$$

$$G = \frac{F/a}{\Delta L/L} = \frac{FL}{a\Delta L}$$

clearly, modulus of rigidity has same SI unit and DF as that of stress. It is involved in solids only.

Q. what is Young's modulus of elasticity for a rigid body?

longitudinal strain for a rigid body -

$$\Delta L = 0$$

$$\therefore LS = \frac{\Delta L}{L} = \frac{0}{L} = 0$$

$$\gamma = \frac{\text{longitudinal stress}}{\text{longitudinal strain}}$$

$$\gamma = \frac{\text{longitudinal stress}}{0}$$

$$\boxed{\gamma = \infty}$$

lateral strain

$$= \frac{\text{change in diameter}}{\text{original diameter}} = \frac{\Delta D}{D}$$

$$\text{or } \frac{\text{change in radius}}{\text{original radius}} = \frac{\Delta r}{r}$$

It is unitless as well as dimensionless.

Q. A wire is stretched by long

Poisson's Ratio (σ)

$$\sigma = \frac{\text{lateral strain}}{\text{longitudinal strain}}$$

$$\sigma = \frac{-\Delta D/D}{\Delta L/L}$$

$$\sigma = \frac{-L \Delta D}{D \Delta L}$$

It is unitless as well as dimensionless.

For practical materials its value lies between 0 and 0.5.

$$0 < \sigma < 0.5$$

✓ Q. A wire of length 2 m and cross sectional area 10^{-6} m^2 is stretched by 0.2 mm by a force of 50 N. Find Young's modulus of elasticity.

$$L = 2 \text{ m.}$$

$$a = 10^{-6} \text{ m}^2.$$

$$\Delta L = 0.2 \text{ mm} = 0.2 \times 10^{-3} \text{ m.}$$

$$F = 50 \text{ N.}$$

$$\text{longitudinal stress} = \frac{F}{a} = \frac{50}{10^{-6}} = 50 \times 10^6 \text{ Pa.} = 5 \times 10^7 \text{ Pa.}$$

$$\text{longitudinal strain} = \frac{\Delta L}{L}$$

$$= \frac{0.2 \times 10^{-3} \text{ m}}{2 \text{ m}} = \frac{2 \times 10^{-4} \text{ m}}{2 \text{ m}} = 10^{-4}.$$

$$Y = \frac{\text{long. stress}}{\text{long strain}}$$

$$Y = \frac{5 \times 10^7}{10^{-4}} = 5 \times 10^{11} \text{ Pa.}$$

- Q A wire of length 2m and diameter 1mm is stretched by 0.2 mm. If diameter decreases by 0.001 mm, find the lateral strain, long strain and Poisson's ratio.

$$l = 2\text{m}$$

$$D = 1\text{mm} = 10^{-3}\text{m}$$

$$\Delta L = 0.2\text{mm} = 0.2 \times 10^{-3}\text{m} = 2 \times 10^{-4}\text{m}$$

$$\Delta D = 0.001\text{mm} = 0.001 \times 10^{-3}\text{m} = 10^{-6}\text{m}$$

$$\text{long strain} = \frac{\Delta L}{L} = \frac{2 \times 10^{-4}}{2} = 10^{-4}$$

$$\text{lateral strain} = \frac{\Delta D}{D} = \frac{10^{-6}}{10^{-3}} = 10^{-3}$$

$$\text{Poisson's ratio} = \frac{\text{lateral strain}}{\text{long strain}} = \frac{10^{-3}}{10^{-4}} = \frac{1}{10^{-1}} = 10^1 = 10$$

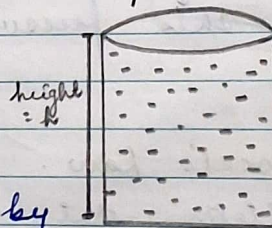
Fluids (liquids and gases).

Pressure exerted by a liquid column.

$$F = mg$$

$$F = \rho V g$$

$$F = \rho A h g$$



Area of base = A

Volume of liquid

$$V = Ah$$

Density = $\frac{m}{V}$

$$\rho = \frac{m}{V}$$

$$m = \rho V$$

Now, Pressure exerted by liquid column

$$P = \frac{F}{A} \rightarrow P = \frac{\rho A h g}{A}$$

$$P = \rho g h$$

effect of gravity on Pressure
 if $g = 0$.
 then, $p = 0$.

i.e; in the absence of gravity, pressure exerted by a liquid column is zero.

In human body, blood pressure is different at different parts.

It is maximum at feet because of the maximum height (h) and it is minimum at head because of the minimum height (h).

Total pressure exerted on the bottom surface:

$$P_T = P + P_a$$

$P_a \rightarrow$ atmospheric pressure.

$$P_a = 1.01 \times 10^5 \text{ Pa}$$

$$P_T = fgh + P_a$$

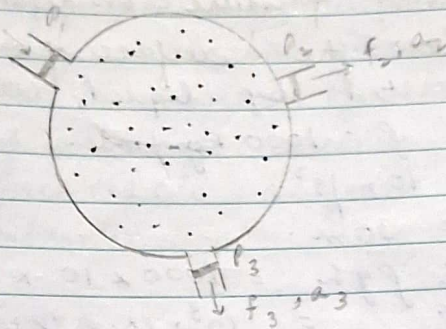
$$\underline{P_T - P_a = fgh}$$

↓
 this pressure difference is known as gauge pressure

Pascal's Law.

It states that the pressure exerted at any point on a liquid in equilibrium in a container is transmitted equally to all the points and to the walls of the container as well.

It is important to note that effect of gravity is to be neglected.



$$P_1 = P_2 = P_3$$

$$\frac{f_1}{a_1} = \frac{f_2}{a_2} = \frac{f_3}{a_3}$$

Applications:

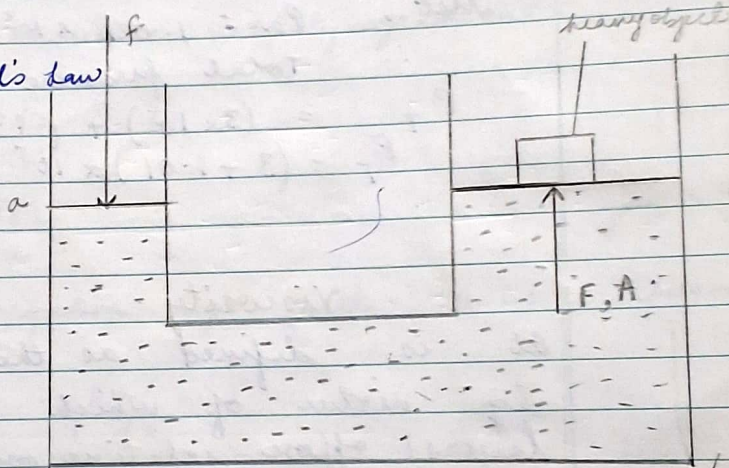
1. Hydraulic lift:

According to Pascal's law

$$\frac{f}{a} = \frac{F}{A}$$

$$\frac{f}{F} = \frac{a}{A}$$

$$\text{If } A > a \\ F > f$$



example:

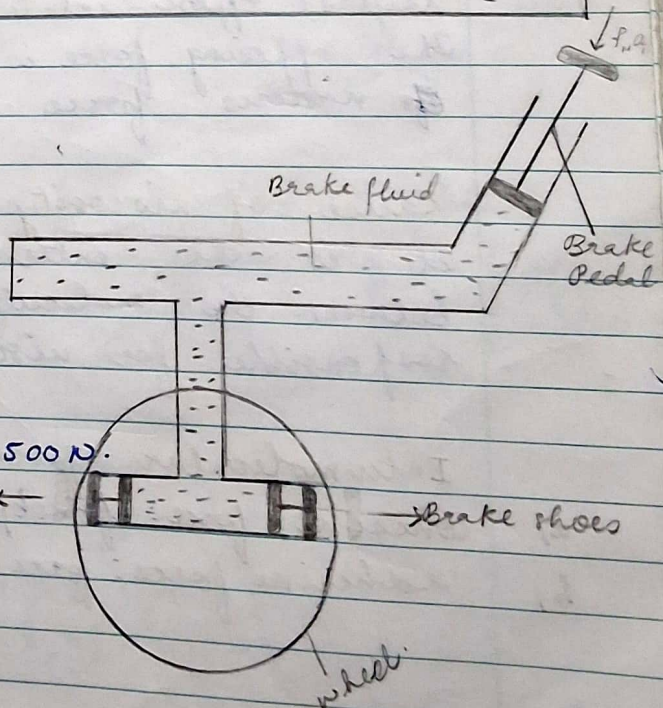
$$a = 2 \text{ m}^2, A = 10 \text{ m}^2$$

$$f = 500 \text{ N}, F = ?$$

$$\frac{500}{F} = \frac{2}{10}$$

$$\frac{500}{F} = \frac{1}{5}$$

$$F = 500 \times 5 = 2500 \text{ N}$$



2. Hydraulic brake:

$$\frac{f_1}{a_1} = \frac{f_2}{a_2} = \frac{f_3}{a_3}$$

Q Find the total pressure on a man at a depth of 30m below the surface of water.
 Pressure exerted by liquid column = ρgh .
 we have, $\rho = 1000 \text{ kg/m}^3$.

$$g = 10 \text{ m/s}^2$$

$$h = 30 \text{ m}$$

$$\therefore P = \rho gh = 1000 \times 10 \times 30$$

$$P = 10^3 \times 10 \times 3 \times 10$$

$$P = 3 \times 10^5 \text{ Pa}$$

$$\text{Also, } P_a = 1.01 \times 10^5 \text{ Pa}$$

$$\therefore \text{Total pressure} = P + P_a$$

$$P_T = (3 \times 10^5) + (1.01 \times 10^5)$$

$$P_T = (3 + 1.01) \times 10^5 = 4.01 \times 10^5 \text{ Pascal}$$

Viscosity.

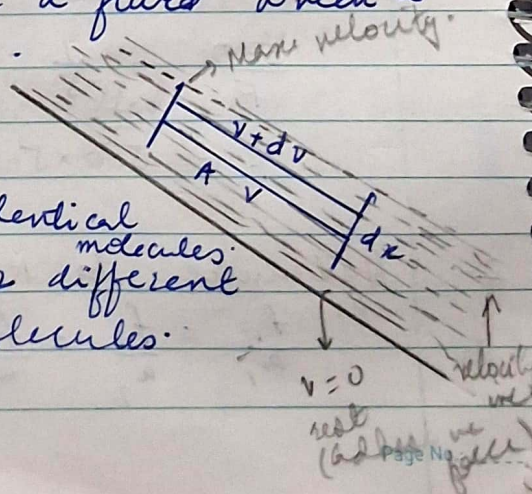
It is defined as the property of a fluid by virtue of which its molecules (molecular layers) oppose relative motion of each other. The opposing force is known as force of viscosity or viscous force.

Cause of viscosity.

It is the intermolecular force of attraction between the molecules of a fluid which is responsible for viscosity.

Intermolecular force

- cohesive force: force b/w 2 identical molecules.
- adhesive force: force b/w 2 different molecules.



- * In case of water in a vessel, adhesive force is stronger than cohesive force, that's why the water wets the walls of the container.
- * In case of mercury, cohesive force is stronger than adhesive force that's why it doesn't wet the walls of a container.

$$F \propto A$$

$$F \propto \frac{dv}{dx}$$

Here, $\frac{dv}{dx}$ is known as velocity gradient.

$$\therefore F \propto A \frac{dv}{dx}$$

$$F = \eta A \frac{dv}{dx}$$

we can put negative sign with it as it represents opposing force.

$$F = -\eta A \frac{dv}{dx}$$

Here η is a constant and is known as coefficient of viscosity.

Different fluids have different values of η (eta).
Honey is more viscous than water. as coefficient of viscosity for honey is greater than that of water.

$$\eta = \frac{F dx}{A dv}$$

$$\text{SI unit} = \frac{\text{Nm}}{\text{m}^2 \text{s}} = \frac{\text{Ns}}{\text{m}^2} = \text{Pa s}.$$

$$\begin{aligned} \text{OF of } \eta &= \text{m}^1 \text{L}^{-1} \text{T}^{-2} \times \text{T}^1 \\ &= \text{m}^1 \text{L}^{-1} \text{T}^{-1} \end{aligned}$$

- In case of liquids, with the increase in temperature viscosity decreases.
Reason: increase in temperature weakens the intermolecular force of attraction.
- In case of gases, with the increase in temperature ^{viscosity} increases.
Reason: Increase in temperature, increases the frequency (no) of collisions.

Types of liquid flow.

1. stream line flow: It is the ordered flow of a liquid and the liquid moves in the form of layers which do not mix with each other. In stream line flow all the molecules passing through a given point possess same velocity at the given point.
It is also known as laminar flow.
2. Turbulent flow: It is the disordered flow of a liquid. In this flow the molecules move randomly and molecular layers mix with each other.
In this flow, all the molecules passing through a given point possess different velocities at the given point.

VV Imp.

Stoke's Law.

This law gives us the viscous force acting on a given body ~~mo~~ (spherical in shape) moving through a fluid.
According to Stoke, the viscous force (F) acting on a spherical body moving through

- a fluid depends on the following factors
- coefficient of viscosity of the fluid (η)
 - radius of the spherical body (r)
 - velocity of the spherical body (v).

$$F \propto \eta^a$$

$$F \propto r^b$$

$$F \propto v^c$$

$$F \propto \eta^a r^b v^c$$

$$F = k \eta^a r^b v^c \quad \text{--- ①}$$

↓
dimensionless constant.

Acc to PHD.

$$M^1 L^1 T^{-2} = (M^1 L^{-1} T^{-1})^a (L^1)^b (L^1 T^{-1})^c$$

$$M^1 L^1 T^{-2} = M^a L^{-a} T^{-a} L^b L^c T^{-c}$$

$$M^1 L^1 T^{-2} = M^a L^{-a+b+c} T^{-a-c}$$

comparing powers.

$$\rightarrow \boxed{a = 1}$$

$$\rightarrow -a - c = -2$$

$$-1 - c = -2$$

$$-1 + 2 = c$$

$$\boxed{1 = c}$$

$$\rightarrow -a + b + c = 1$$

$$-1 + b + 1 = 1$$

$$\boxed{b = 1}$$

From eqn. ①.

$$F = k \eta^1 r^1 v^1$$

$$F = k \eta r v$$

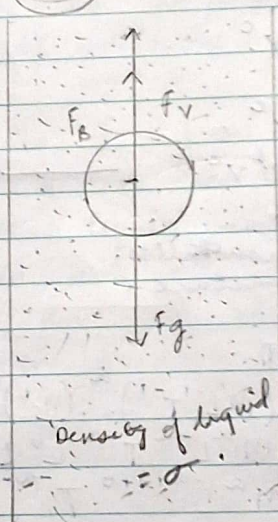
$$\text{but } k = 6\pi$$

$$\therefore F = 6\pi \eta r v$$

Terminal velocity (v_T)

it is defined as the maximum constant velocity attained by a body falling through a fluid when the net force on it becomes zero.

$$\rho \quad \text{---} \quad m, \frac{4}{3}\pi r^3$$



1 Forces acting on the body
weight of the body

$$F_g = mg$$

$$F = \frac{4}{3}\pi r^3 \rho g \quad \text{Downwards}$$

2 viscous force

$$F_v = 6\pi \eta r v \quad \text{upwards}$$

3 Buoyant force

$$F_B = \text{weight of the fluid displaced}$$

$$F_B = \frac{4}{3}\pi r^3 \sigma g \quad \text{upwards}$$

During the fall of a body through a fluid a stage comes where the net force acting on the body becomes zero (apparent weight of the body becomes zero).
i.e; the body attains constant velocity. This constant velocity is known as terminal velocity.

$$F_g = F_v + F_b$$

$$\frac{4}{3} \pi r^3 \rho g = 6 \pi \eta r v_T + \frac{4}{3} \pi r^3 \sigma g$$

$$6 \pi \eta r v_T = \frac{4}{3} \pi r^3 \rho g - \frac{4}{3} \pi r^3 \sigma g$$

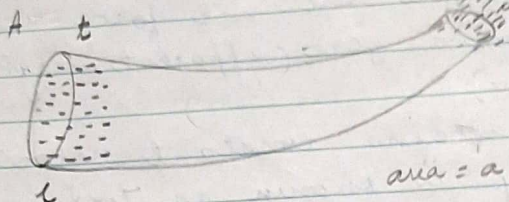
$$6 \pi \eta r v_T = \frac{4}{3} \pi r^3 (\rho - \sigma) g$$

$$6 \eta v_T = \frac{4}{3} r^2 (\rho - \sigma) g$$

$$v_T = \frac{4 r^2 (\rho - \sigma) g}{18 \eta}$$

$$v_T = \frac{2 r^2 (\rho - \sigma) g}{9 \eta}$$

Equation of continuity.



$$\begin{aligned} \text{area} &= a_1 \\ \text{velocity} &= v_1 \\ \text{density} &= \rho_1 \end{aligned}$$

$$\begin{aligned} \text{area} &= a_2 \\ \text{velocity} &= v_2 \\ \text{density} &= \rho_2 \end{aligned}$$

Ideal fluid

- 1) viscosity is negligible.
- 2) it is incompressible.

Mass of the liquid entering at cross-section A per second:
 $= a_1 v_1 \rho_1$

$$M = \rho V$$

$$M = \rho a_1 \times l$$

$$\text{mass} = \rho a_1 \left(\frac{l}{t} \right) v_1$$

Mass of the liquid leaving at cross section B per second:
 $= a_2 v_2 \rho_2$

$$\therefore M/s \quad a_1 v_1 \rho_1$$

Acc to conservation of mass.

$$a_1 v_1 \rho_1 = a_2 v_2 \rho_2$$

If the fluid is incompressible.
 $\therefore \rho_1 = \rho_2 = \rho$ (say)

$$\therefore a_1 v_1 \cancel{\rho} = a_2 v_2 \cancel{\rho}$$

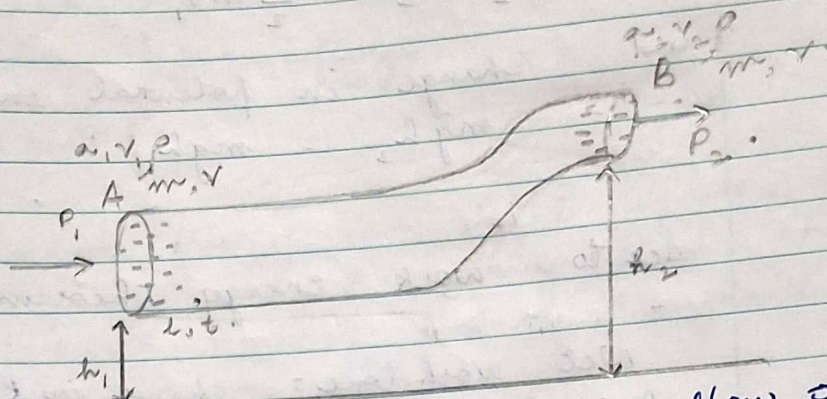
$$\therefore a_1 v_1 = a_2 v_2$$

Equation of continuity actually represents Law of conservation of mass.

$$\text{As } a_1 > a_2$$

$$\Rightarrow v_1 < v_2$$

On general $av = \text{constant}$.

Bernoulli's Theorem.

It states that in a stream line flow of an ideal liquid (non-viscous, incompressible) the sum total of its pressure energy per unit volume, kinetic energy and potential energy per unit volume remains constant at every cross-section.

OR

The sum total of pressure energy per unit mass, kinetic energy and potential energy per unit mass remains constant at every cross-section.

Work done per second at cross-section A on the liquid.

$$W_1 = P_1 V$$

Work done per second at cross-section B by the liquid

$$W_2 = P_2 V$$

$$a_1 v_1 \rho = a_2 v_2 \rho = m$$

$$a_1 v_1 = a_2 v_2 = \frac{m}{\rho} = V$$

$$(\rho = \frac{m}{V})$$

$$W = f \cdot l$$

$$W = \frac{f}{a} \frac{al}{t} = P V$$

$$\text{Net work done} = W_1 - W_2 = P_1 V - P_2 V$$

Now, change in kinetic energy per second
 $= \frac{1}{2} m v_2^2 - \frac{1}{2} m v_1^2$

and change in potential energy per second
 $= m g h_2 - m g h_1$

According to work energy Theorem.

Net work done = change in K.E + Change in P.E.
 $P_1 V - P_2 V = \frac{1}{2} m v_2^2 - \frac{1}{2} m v_1^2 + m g h_2 - m g h_1$ *

$$P_1 V + \frac{1}{2} m v_1^2 + m g h_1 = P_2 V + \frac{1}{2} m v_2^2 + m g h_2$$

Pressure energy.

Dividing b/s by "V"

Pressure energy per unit volume

$$P_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2$$

Dividing b/s by "m"

$$P_1 V + \frac{1}{2} m v_1^2 + m g h_1 = P_2 V + \frac{1}{2} m v_2^2 + m g h_2$$

$$\frac{P_1 V}{m} + \frac{1}{2} v_1^2 + g h_1 = \frac{P_2 V}{m} + \frac{1}{2} v_2^2 + g h_2$$

$$\frac{P_1}{\rho} + \frac{v_1^2}{2} + g h_1 = \frac{P_2}{\rho} + \frac{v_2^2}{2} + g h_2$$

In general,

$$\frac{P}{\rho} + \frac{v^2}{2} + g h = \text{constant.}$$

Dividing b/s by 'g'

$$\frac{P}{\rho g} + \frac{v^2}{2g} + h = \text{constant}$$

$\downarrow$ Pressure head $\downarrow$ velocity head $\downarrow$ gravitational head

In a streamline flow of an ideal liquid, the sum of its pressure head, velocity head and gravitational head remains constant throughout its flow at every cross-section.

* If both ends are at same height.

$$\frac{P_1}{\rho} + \frac{v_1^2}{2} = \frac{P_2}{\rho} + \frac{v_2^2}{2}$$

In general

$$\frac{P}{\rho} + \frac{v^2}{2} = \text{constant}$$

Dividing b/s by 'g'

$$\frac{P}{2\rho} + \frac{v^2}{2g} = \text{constant}$$

clearly, high pressure implies low velocity and low pressure implies high velocity.

Critical velocity (V_c)

It is defined as the maximum velocity of a liquid below which its flow is stream lines and above which its flow becomes turbulent.

→ critical velocity (V_c) of a liquid depends on its coefficient of viscosity (η), its density (ρ) and the diameter of the pipe (D).

$$V_c \propto \eta$$

$$V_c \propto \frac{1}{\rho}$$

$$V_c \propto \frac{1}{D}$$

$$V_c \propto \frac{\eta}{\rho D}$$

$$V_c = \frac{R\eta}{\rho D}$$

R is a constant and is known as Reynolds's no.

Reynolds's no. is unitless as well as dimensionless.

For streamline flow $R < 2000$.

For turbulent flow $R > 3000$.

If R lies b/w 2000 and 3000, flow is said to be unstable.

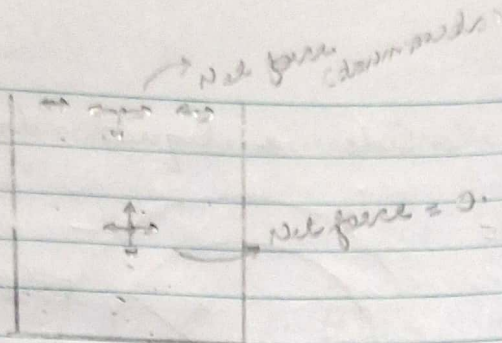
Surface Tension.

It is defined as the property of a liquid by virtue of which the free surface of a liquid behaves as an elastic membrane and tends to have a minimum surface area.

- With the increase in temperature surface tension of a liquid decreases.

- A needle floats on the surface of water due to surface tension.

- An insect crawls on the surface of water due to surface tension.



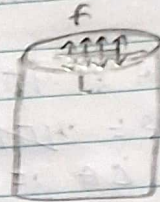
Raindrops are spherical in shape because spherical shape has minimum surface area and is most stable. Same applies to planets.

$$\text{Surface Tension } T = \frac{F}{l}$$

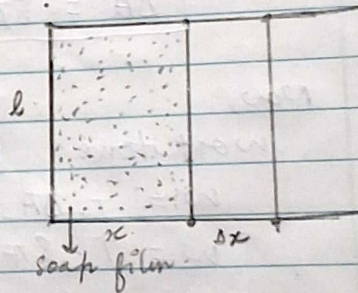
$$\text{SI unit of } T = \text{N/m}$$

$$\text{OF of } T = \frac{M^1 L^1 T^{-2}}{L^1}$$

$$= M^1 T^{-2}$$



Surface Energy.
Minimum force to be applied
 $F = 2Jl$



Now,

workdone in increasing area

$$W = F \Delta x$$

$$W = 2Jl \Delta x$$

$$W = J(2l \Delta x)$$

$$W = J \Delta A$$

inc in area.

This workdone is stored in the form of potential energy in the surface molecules and is known as surface energy.

$$\therefore E = J \Delta A$$

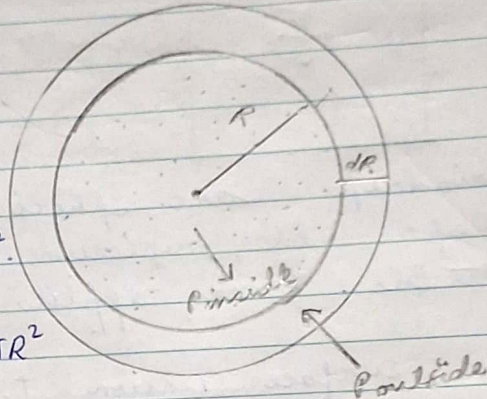
$$\text{Surface energy} = \frac{\text{Surface}}{\text{Increase in Area}} \times \text{Increase in Area}$$

Excess Pressure inside a liquid drop

Excess pressure $P_0 =$
 $P_{\text{inside}} - P_{\text{outside}}$

Increase in surface
 area = $4\pi(R + dR)^2 - 4\pi R^2$

$$\Delta A = 4\pi(R + dR)^2 - 4\pi R^2$$



$$\Delta A = 4\pi(R^2 + dR^2 + 2RdR) - 4\pi R^2$$

$$\Delta A = \cancel{4\pi R^2} + 4\pi dR^2 + 8\pi R dR - \cancel{4\pi R^2}$$

$$\Delta A = 4\pi dR^2 + 8\pi R dR$$

neglecting dR^2 term

$$\Delta A = 8\pi R dR$$

Now,

work done

$$W = T \cdot \Delta A$$

$$W = T \cdot 8\pi R dR \quad \text{--- (1)}$$

Also

$$W = P_0 4\pi R^2 dR \quad \text{--- (2)}$$

$$W = F dR$$

$$W = P \Delta A$$

comparing eq (1) and (2)

$$P_0 4\pi R^2 dR = T 8\pi R dR$$

$$P_0 = \frac{8T}{4R}$$

$$P_0 = \frac{2T}{R}$$

Excess Pressure inside a soap bubble

In this case two surfaces are involved.

Increase in surface Area.

$$\Delta A = 2 \left[4\pi (R+dR)^2 - 4\pi R^2 \right]$$

$$\Delta A = 8\pi (R^2 + dR^2 + 2RdR) - 8\pi R^2$$

$$\Delta A = 8\pi R^2 + 8\pi dR^2 + 16\pi RdR - 8\pi R^2$$

$$\Delta A = 8\pi dR^2 + 16\pi RdR$$

Neglecting dR^2 term

$$\Delta A = 16\pi RdR$$

Now,

work done

$$W = T \cdot \Delta A$$

$$W = T \cdot 16\pi RdR \quad \text{--- (1)}$$

Also

$$W = P_0 \cdot 4\pi R^2 dR \quad \text{--- (2)}$$

comparing eq (1) and (2)

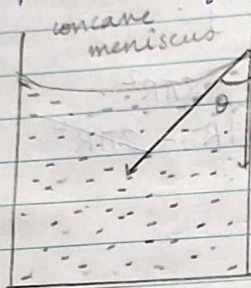
$$P_0 \cdot 4\pi R^2 dR = T \cdot 16\pi RdR$$

$$P_0 = \frac{4T}{R}$$

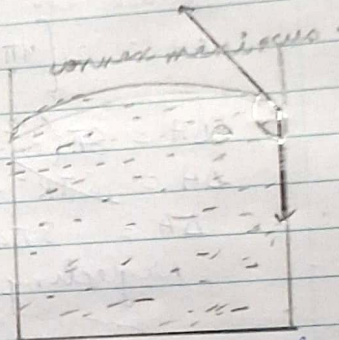
$$P_0 = \frac{4T}{R}$$

V.V.Smp. Angle of contact.

It is defined as the angle b/w the tangent to the liquid surface and the tangent to the solid surface within the liquid at the point of contact.



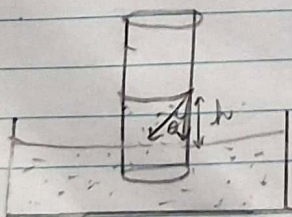
- liquid which wets the walls of a container (e.g.; water).
- cohesive force > adhesive force
- Angle of contact is acute ($< 90^\circ$).



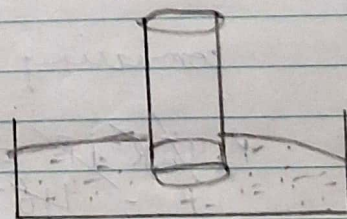
- liquid which doesn't wet the walls of a container (e.g.; mercury)
- Adhesive force < cohesive force.
- angle of contact is obtuse ($> 90^\circ$).

Capillarity.

Capillary tube: A tube with a fine bore.



WATER



MERCURY.

The rise or fall in the level of a liquid in a capillary tube when it is placed in a liquid is known as capillarity.

If the liquid is water, the level rises

If the liquid is mercury, the level falls.

Ascent formula or Jurin's formula
This formula gives us the height upto which a liquid can rise up in a capillary tube.

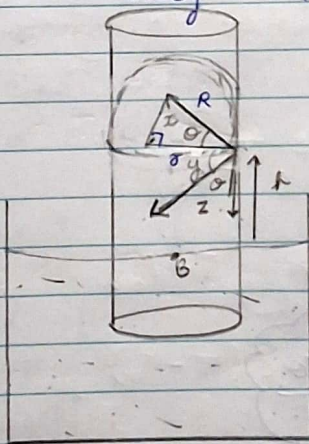
If 'T' is the surface tension of the liquid, 'θ' is the angle of contact, 'r' is the radius of the capillary tube, 'ρ' is the density of the liquid.

Then.

$$h = \frac{2T \cos \theta}{r \rho g}$$

For water $\theta < 90^\circ$, h is positive (level rises).

For mercury $\theta > 90^\circ$, h is negative (level decreases falls / -ve).



$$x + z = 90^\circ$$

$$x + y = 90^\circ$$

$$y + z = x + y$$

$$z = x.$$

$$h = \frac{2T \cos \theta}{r \rho g}$$

(r - radius of tube.)

Proof: At bottom point B.

$$P_B = \rho g h.$$

at point A

$$P_A = \frac{2T}{R} \quad (R - \text{radius of meniscus}).$$

at equilibrium.

$$P_B = P_A$$

$$\rho g h = \frac{2T}{R}$$

$$h = \frac{2T}{R \rho g}$$

we have,

$$\cos \theta = \frac{r}{R}$$

$$R = \frac{r}{\cos \theta}$$

$$h = \frac{2T}{\frac{r}{\cos \theta} \rho g}$$

$$h = \frac{2T \cos \theta}{r \rho g}$$

Thermal expansion.

The increase in the dimensions of a body (length, area or volume) when heat is given to it i.e; when its temperature is increased is known as thermal expansion.

Types:

- 1) Linear expansion: The increase in the length of a body due to increase in its temperature is known as linear expansion.

 L, T $T + \Delta T$ $L + \Delta L$

$$\Delta L \propto L$$

$$\Delta L \propto \Delta T$$

$$\Delta L \propto \Delta T$$

$$\Delta L = \alpha L \Delta T$$

α is a constant and is known as coefficient of linear expansion.

$$\alpha = \frac{\Delta L}{L \Delta T}$$

SI unit of $\alpha = \frac{m}{m \cdot K} = K^{-1}$

value of α is very very small.

Initial length = L_0 , initial temp = T_0 .
Final length = L , final " = T .

$$L - L_0 = \alpha L_0 (T - T_0)$$

$$L = L_0 + \alpha L_0 (T - T_0)$$

$$L = L_0 [1 + \alpha (T - T_0)]$$

2. Aerial expansion: The increase in the area of a body due to increase in its temperature is known as aerial expansion or superficial expansion.

$$A, T.$$

$$A + \Delta A, T + \Delta T.$$

$$\Delta A \propto A$$

$$\Delta A \propto \Delta T$$

$$\Delta A \propto A \Delta T$$

$$\Delta A = \beta A \Delta T$$

β is a constant and is known as the coefficient of aerial expansion.

S.I unit of $\beta = K^{-1}$

Initial area = A_0 , Initial temp = T_0

Final area = A , Final temp = T

$$A - A_0 = \beta A_0 (T - T_0)$$

$$A = A_0 + \beta A_0 (T - T_0)$$

$$A = A_0 [1 + \beta (T - T_0)]$$

3. Volume expansion: The increase in the volume of a body due to increase in its temperature is known as volume expansion.



$$V, T$$

$$V + \Delta V, T + \Delta T$$

$$\Delta V \propto V$$

$$\Delta V \propto \Delta T$$

$$\Delta V \propto V \Delta T$$

$$\Delta V = \gamma V \Delta T$$

Here γ is a constant and is known as coefficient of volume expansion.

SI unit of γ is K^{-1} .

very very small

Initial volume = V_0 , Initial temp = T_0

Final volume = V , Final temp = T .

$$V - V_0 = \gamma V_0 (T - T_0)$$

$$V = V_0 + \gamma V_0 (T - T_0)$$

$$V = V_0 [1 + \gamma (T - T_0)]$$

v^o imp.

Relation between α and β .

we have,

$$A + \Delta A = (L + \Delta L)^2$$

$$A + \beta A \Delta T = (L + \alpha L \Delta T)^2$$

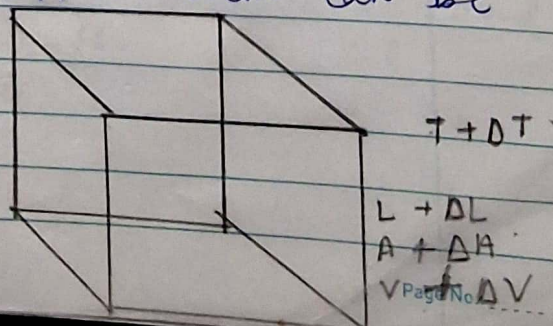
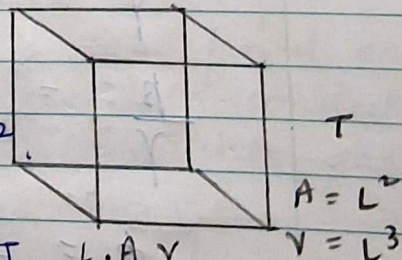
$$A (1 + \beta \Delta T) = L^2 (1 + \alpha \Delta T)^2$$

$$1 + \beta \Delta T = 1 + \alpha^2 \Delta T^2 + 2\alpha \Delta T$$

As α is very very small, $\therefore \alpha^2$ term can be neglected.

$$\therefore \beta \Delta T = 2\alpha \Delta T$$

$$\beta = 2\alpha$$



Relation between α and γ

we have,

$$V + \Delta V = (L + \Delta L)^3$$

$$V + \gamma V \Delta T = (L + \alpha L \Delta T)^3$$

$$\cancel{V} (1 + \gamma \Delta T) = \cancel{L^3} (1 + \alpha \Delta T)^3$$

$$\cancel{V} + \gamma \Delta T = \cancel{V} + \alpha^3 \Delta T^3 + 3\alpha \Delta T + 3\alpha^2 \Delta T^2$$

$$\gamma \Delta T = \alpha^3 \Delta T^3 + 3\alpha \Delta T + 3\alpha^2 \Delta T^2$$

As α is very very small, $\therefore$ neglecting α^2 and α^3 terms.

$$\gamma \cancel{\Delta T} = 3\alpha \cancel{\Delta T}$$

$$\gamma = 3\alpha$$

Relation between β and γ

we have.

$$\beta = 2\alpha$$

$$\gamma = 3\alpha$$

$$\frac{\beta}{\gamma} = \frac{2\cancel{\alpha}}{3\cancel{\alpha}}$$

$$\beta = \frac{2}{3} \gamma$$

Anomalous expansion of water

Generally, with the increase in temperature a body expands and with the decrease in temperature it contracts, but water shows different behaviour. When the temperature of water is increased from 0°C to 4°C , it contracts or we can say when the temperature of water is decreased from 4°C to 0°C , it expands.

NOTE: Density of water is maximum at 4°C .

Latent Heat:

Also known as hidden heat.

It is defined as the amount of heat required to convert a body of mass 1 kg (or 1 g) from one state to another state without any change in its temperature.

- Latent heat of fusion: It is defined as the amount of heat required to convert 1 kg (or 1 g) of a solid substance into liquid form at its melting point.
- Latent heat of vaporisation: It is defined as the amount of heat required to convert 1 kg (or 1 g) of a liquid substance into gaseous form at its boiling point.

If ' m ' is the mass of the substance, and ' L ' is its latent heat then the total heat is given by:

$$Q = mL$$

Modes of heat transfer.

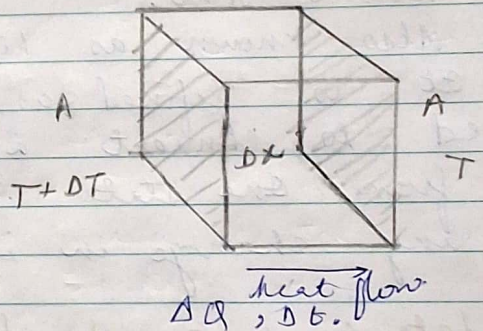
1. conduction: In this mode of heat transfer, heat is transferred from one point to another point without any actual movement of particles of the medium between the two points.
In this mode heat gets transferred from particle to particle.
All solids get heated by this mode.

Rate of heat flow,

$$\frac{\Delta Q}{\Delta t} \propto A$$

$$\frac{\Delta Q}{\Delta t} \propto \Delta T$$

$$\frac{\Delta Q}{\Delta t} \propto \frac{1}{\Delta x}$$



$$\frac{\Delta Q}{\Delta t} \propto \frac{A \Delta T}{\Delta x}$$

$$\frac{\Delta Q}{\Delta t} = \frac{k A \Delta T}{\Delta x}$$

where k is a constant and is known as coefficient of thermal conductivity / thermal conductance.

2 **Convection:** In this mode, heat is transferred from one point to another point due to actual movement of particles of the medium between the two points. (except mercury)
 All liquids and gases get heated by convection. Mercury is heated by conduction.

3 **Radiation:** In this mode heat is transferred in the form of radiations. This mode doesn't require any medium for transferring heat from one point to another point.

Black body radiation.

A black body is a body which absorbs all the wavelengths of all colours of light incident on it.

A black body doesn't reflect any light that is why it appears black. as black means no colour.

When a black body is heated it emits radiations of different wavelengths. These radiations are known as black body radiations.

LAWS

(1) **Stefan - Boltzman's law.**
 (Energy emitted per unit area per unit time) E

$\propto T^4$ or T^4

highest E

lowest E

It states that the energy emitted per unit area per second by a black body when heated is directly proportional to 4th power of its temperature.

highest frequency

lowest frequency

shortest wavelength

longest wavelength

$$E \propto T^4$$

$$E = \sigma T^4$$

σ is called Stefan's constant.

$$\sigma = \frac{E}{T^4}$$

SI unit of $\sigma = \frac{\text{J m}^{-2} \text{s}^{-1}}{\text{K}^4} = \text{J m}^{-2} \text{s}^{-1} \text{K}^{-4}$

$= \text{Watt m}^2 \text{K}^{-4}$

(Power $= \frac{E}{t} = \text{J s}^{-1}$)

$$\sigma = 5.67 \times 10^8 \text{ watt m}^2 \text{K}^{-4}$$

$$\sigma = \frac{E}{T^4}$$

$$= \frac{\text{Energy}}{\text{area} \times \text{time}}$$

$$\frac{\text{Joule}}{\text{m}^2 \text{sec}} = \text{J m}^{-2} \text{sec}^{-1}$$

② Wein's displacement law.

The wavelength of the radiation emitted by a black body when heated is inversely proportional to its temperature.

$$\lambda \propto \frac{1}{T}$$

$$\lambda = \frac{b}{T}$$

b is known as weins constant.

$$b = 2.898 \times 10^{-3} \text{ mK}$$

- Q. A black body at 3000 K has peak at 966 nm.
 A. what λ (wavelength) will it peak at 6000 K?
 we have

$$\lambda = \frac{b}{T}$$

$$\lambda T = b.$$

now, $966 \text{ nm} \times 3000 = b.$
 $\lambda \times 6000 = b.$

$$\therefore 966 \times 3000 = \lambda \times 6000$$

$$\lambda = \frac{966 \times 3000}{6000} = \frac{966}{2} = 483 \text{ nm}$$

- Q. The sun emits radiation of wavelength 480 nm.
 Find its surface temperature.

$$\lambda = 480 \text{ nm} = 480 \times 10^{-9} \text{ m}.$$

$$\lambda = \frac{b}{T}$$

$$T = \frac{b}{\lambda}$$

$$T = \frac{2.898 \times 10^{-3}}{480 \times 10^{-9}}$$

$$T = \frac{2898 \times 10^6}{480 \times 10^3} = \frac{2898}{480} \times 10^3.$$

$$= 6 \times 10^3 = 6000 \text{ K}.$$

- Q. Find the excess pressure inside a water drop of radius 1 mm. (Take $\gamma = 0.072 \text{ N/m}$)

$$P = \frac{2\gamma}{R}$$

$$1 \text{ mm} = 10^{-3} \text{ m}$$

$$P = \frac{2 \times 0.072}{10^{-3}}$$

$$P = 0.144 \times 10^3$$

$$P = \frac{144}{10^3} \times 10^3$$

$$P = 144 \text{ Pa}$$

- Q. A small sphere of radius 2 mm falls through water with a velocity of 2 cm/s. Find the viscous force acting on the sphere if the coefficient of viscosity for water is $10^{-3} \text{ Pa}\cdot\text{s}$.

$$F_v = 6\pi\eta r v$$

$$\text{radius} = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$$

$$\text{velocity} = 2 \text{ cm/s} = 2 \times 10^{-2} \text{ m/s}$$

$$\therefore F_v = \frac{6 \times 22}{7} \times 10^{-3} \times 2 \times 10^{-3} \times 2 \times 10^{-2}$$

$$F_v = 75.36 \times 10^{-8} \text{ N}$$

Calorimetry.

It is the branch of physics which deals with the study of measurement of heat. The device which is used to measure the heat present in a system is known as calorimeter.

Principle:

Heat lost by one system is equal to heat gained by the other system.

→ If 'C' is the specific heat of a substance, 'm' is its mass and 'T' is its temperature then the heat present in the substance is given by:

$$Q = mCT.$$

Heat lost = Heat gained

$$m_1 C_1 T_1 - m_1 C_1 T = m_2 C_2 T - m_2 C_2 T_2$$

$$m_1 C_1 (T_1 - T) = m_2 C_2 (T - T_2)$$

Q. 50g of water at temperature 30°C is mixed with 100g of water at 50°C. Find the final temperature.

$$50 \times (30 - T) = 100 \times (T - 50)$$

$$30 - T = 2(T - 50)$$

$$30 - T = 2T - 100$$

$$30 + 100 = 2T + T$$

$$3T = 130$$

$$T = \frac{130}{3} = 43.3^{\circ}\text{C}$$